

PART I AN OVERVIEW

Chapter I "The Energy Crisis: The Problem & A Possible Solution"

As we end living in the eighth decade of the twentieth century, we have discovered a reality that until the beginning of this decade seemed a sublime dream - the so-called "energy crisis". At the present rate of depletion of the world's energy reserves, the world by the end of this century will be running on borrowed "energy time". Moreover, many leading meteorologists now believe that the climate of the planet and especially the northern hemisphere could be drastically altered if we continue to burn fossil fuels for the two remaining decades of this century. Indeed, it now appears that the planetary civilization that was built on the abundance of fossil fuels such as coal, oil, and natural gas is rapidly coming to an end. The old assumptions are no longer valid and new truths must come forth to solve the situation if civilization as we now knew it is to continue. In this light, the amazing insights and technology of a presently obscure inventor and philosopher named Nikola Tesla appear even more than ever before to offer the new truths that are capable of allowing humanity the methodology that could well save our planet by solving the energy crisis that at present seems beyond the human experience. To explore this possibility, let us begin by observing the true extent of the energy crisis and end with a brief overview of Tesla's solution to the matter.

The Crisis in Energy

If one is to truly gauge the extent of the energy crisis, it is wise to review the amount of energy consumption, the possible reserves of fossil fuels and alternative technology that is presently available to us. In this regard, let us first look at the consumption of fossil fuels. In this context, fossil fuels will be defined as coal, natural gas, and oil products that are employed to heat and light our homes, offices, farms and factories. At the beginning of this decade, we consumed about 21.5 C of energy (see table 1.1)

TABLE 1.1 SOURCES OF ENERGY FOR THE WORLD IN 1970

Source	Energy Content	
	C(= 10^{16} Btu)	%
Fossil fuel		
Coal	6.5	30
Petroleum	7.7	36
Natural Gas	3.8	18

TABLE 1.1 SOURCES OF ENERGY FOR THE WORLD IN 1970 (Continued)

Source	Energy Content	
	C(=10 ¹⁶ Btu)	%
Solar Energy		
Hydroelectricity	1.3	6
Traditional		
Wood, waste, feed	2.2	10
	21.5	100

(Source: J.C. Fisher, Energy Crisis in Perspective (New York, John Wiley & Sons, Inc., 1974), p.16)

Through this decade, the world wide consumption of energy has increased at the level of about five to seven per cent per year. Hence, if we take the world consumption level to be 21.5 C in 1970 then the consumption level will increase to about 34.5 C in 1980. By the end of this century if the increase were constant, the amount of world energy consumption would have increased to about 60.3 C or nearly triple the usage of energy in 1970! However, the actual usage of energy will probably be much higher when the world consumption rate is measured against that of the United States. This comparison can be best made from the following table.

TABLE 1.2 SOURCES OF ENERGY FOR THE UNITED STATES IN 1970

Source	Energy Content	
	C(=10 ¹⁶ BTU)	%
Fossil Fuel		
Coal	1.28	20
Petroleum	2.65	41
Natural Gas	2.13	33

TABLE 1.2 SOURCES OF ENERGY FOR THE UNITED STATES IN 1970 (Continued)

Source	Energy Content	
	C($=10^{16}$ Btu)	%
Solar Energy		
Hydroelectricity	0.26	4
Other		
Miscellaneous	0.13	2
	6.45	100

(Source: J.C. Fisher, Energy Crisis in Perspective (New York, John Wiley & Sons, Inc., 1974) , p.15)

The table demonstrates two alarming points. First, the United States consumption level is about thirty per cent of the world level of consumption. In the beginning of the previous decade, the United States accounted for about forty per cent of the consumption of energy. The rapid shrinking of the energy dominance of the United States demonstrates the beginning of industrialization in areas of the globe that were formerly agricultural. A second point is centered around the difference in the energy mix between the world and the United States. In general, the world in 1970 employed around fifty-four per cent of its total as petroleum and natural gas while the United States, the most industrialized, employed around seventy-four per cent of its total as petroleum and natural gas. Hence, we seem to be not only in the midst of a exponential rise in the consumption of energy but also an especially high useage of petroleum and natural gas. To explore the full meaning of this problem, let us now look at the known reserves of fossil fuels and especially that of oil and natural gas. In this regard, the following table gives a picture of our present reserves of fossil fuels.

TABLE 1.3 RESERVES OF FOSSIL FUELS IN THE UNITED STATES IN 1970

Source	Energy Content	
	C($=10^{16}$ Btu)	C($=10^{16}$ Btu)
	Recoverable	Undiscovered
Natural Gas	48	620

TABLE 1.3 RESERVES OF FOSSIL FUELS IN THE UNITED STATES IN 1970
(Continued)

Source	Energy Content	
	C($=10^{16}$ Btu)	C($=10^{16}$ Btu)
	Recoverable	Undiscovered
Crude Oil	200	270
Coal	3700	3800
Shale Oil	90	800
	4038	5490

(Source: J.C. Fisher, Energy Crisis in Perspective (New York, John Wiley & Sons, Inc., 1974) , pp. 30-32)

As we can see from the table, the vast majority of known reserves in the United States consists of coal deposits. Yet coal in 1970 accounted for only twenty per cent of the total energy consumption. Meanwhile natural gas and crude oil which together accounted for seventy-four per cent of the total energy consumption in the United States comprise only six per cent of recoverable reserves and eighteen per cent of undiscovered reserves. Clearly, the United States was approaching a crisis in the rate and mix of its energy needs versus the amount of domestic reserves. This discrepancy created the danger that surfaced in 1973 during the Arab oil embargo. In fact by 1976, the foreign crude percentage had reached about eighteen per cent of the total amount of crude oil used in the United States. This fact was the major one that caused President Carter to view this problem as one that required the same policy as that required by war.

A further example of the problem at hand was first addressed to the planet by the famous study by Donella and Dennis Meadows, et. al. entitled Limits To Growth. First published in 1968 and later reprinted in numerous editions, this work was the first book to call attention to the immediate dangers of unlimited growth and rapid industrialization in the third world. In it, the M.I.T. research team stated that the planet would run out of crude oil and natural gas by the middle of the next century and that even if coal were then utilized, the fossil fuel reserves would only last for the next few centuries at best. In addition to this dire warning, there is the recent disclosure by many leading meteorologist that the continued useage of fossil fuels could cause a rapid change in the climate of the northern hemisphere and other parts of the globe by the end of this century. Clearly, there must be developed a means to provide a clean (non-polluting) alternative to fossil fuels. At this point, solar and nuclear fuels are usually

offered as such viable alternatives. However, nuclear fuels possess a severe health hazard while solar energy as presently envisioned in the form of collectors and other types of units such as mirrors which reflect their energy on to steam generation units are presently highly unefficient. Indeed, the dangers of nuclear power plants are not only in the as yet unsolved problem of the disposal of nuclear wastes, but also in the danger of thermal pollution to the lakes, rivers or streams that surround such power plants. An additional problem involves the utilization of so-called "breeder reactors" whose output in the form of enriched plutonium can be used to produce atomic weapons. In response to this problem, several nuclear research groups announced that they had created a special type of "breeder reactor" whose waste materials could not be used for the production of nuclear weapons. Regardless of this important technological breakthrough, it still remains a fact that the waste materials from the "breeder reactor" were more dangerous than the conventional reactor since its waste product takes longer to decompose (higher radioactive halflife). The only other resource available to present technology are the various types of wind generators and thermal energy generators. This device as well has only a limited useage as it is tied to those areas that possess large thermal springs and/or a relatively large amount of high speed winds in their vicinity.

Due to the above mentioned difficulties, it is quite necessary for the planet to accept at this time a solution to the above mentioned technical difficulties by adopting a solution first worked out by Nikola Tesla in the first years of this century. In so doing, it becomes possible to adopt a methodology that is capable of providing a end to the energy morass that we find ourself in.

The Tesla "Multifaceted Solution"

In the last decade of the nineteenth century, Nikola Tesla developed the technology to solve the present energy crisis as well as the approach to be followed to best utilize the master wizard's inventions. The key to Tesla's system is divided into three major parts. The first of these is the invention in 1899 of the "magnifying transmitter" and its corollary, the wireless power system. The next major component of the solution is the article in Century Magazine entitled "The Increasing of Human Energy". The final part of the solution involves the patents on the "radiant energy" machines that are taken out in 1901. In order to explain as completely as possible the energy solution of Tesla, let us review each aspect of the system as completely as is now possible.

To begin with, let us look at the "wireless power system". This system consisted of a specially modified Tesla "oscillator" called a "magnifying transmitter" that broadcast electrical power through the earth and the resonance that it set up was received on any point of the globe by use of another Tesla "oscillator" that was modified to pick up the unique signal sent to it by the "magnifying transmitter". In this way Tesla was able to send enough electrical power to light 200 sixty watt lightbulbs a distance of twenty-six miles from the transmitter! The scientific basis for the success of the device lay in the fact that Tesla did not use electric energy in the standard way. Instead, Tesla attempted to use electric energy in a way analogous to a pump. The electrical master believed that electricity was in some way re-

lated to what in classical physics was called an ether. This ether was supposed to be located between the various planets, and the heavenly bodies were supposed to be floating in an energy sea comprised of this ether. Ether was supposed to have mass and to be a non-compressible "fluid". Since the Michaelson-Morley experiment's interferometer was unable to perceive any drag due to an ether, it was quickly assumed by most physicist that the concept of an ether was disproven. In fact, the experiment only tended to disprove the classical contention that ether possessed "mass". On the whole, it did not disprove the entire concept of a massless ether or a unique kind of vacuum energy. In recent years, the new theories espoused by Quantum Physics and especially, the many worlds theories of Everett, Wheeler and others tended to validate such a hypothesis. This energy due to the high vacuum state of its point of origin gave this energy its name of "zero point" energy. There are several points in Tesla's notes that seem to indicate that this type of energy, that modern theorist have found to be electrical in nature, proved a major contention of Tesla that all energy in the universe is in some manner a part of electricity. Tesla believed in this theory whole-heartedly and was convinced that the electrical vibration of the planet by his extremely large high voltage coils was able to successfully transport an immense amount of electrical energy from the transmitter to any point on the globe. To those scientist that might doubt this contention due to the large resistances encountered just in the immediate surface area of the earth, Tesla would have two replies. First, the effect was a synergistic one. That is, the resonances produced by the "magnifying transmitter" overcame this problem since the planet when set at a specific electric resonance acted alot differently then when a high voltage charge was pumped into a unresonanced planet. Finally, the nature of the infinite ground utilized in the oscillator circuit must be taken into account. If Tesla was right about the underlying nature of all things to be electrical and if the electrical vibration was induced through the electrode ball then the concept of Tesla was sound. The principle would much the same as that of the use of an infinite ground in a speaker system to keep the sound (resonances) from attenuating (distorting) too rapidly and causing the system to incorrectly reproduce sound. Tesla, therefore, used electricity differently from the present conventional theories and evidently believed as did Ehrenheft in the fact that the basic unit of charge was not the electron. When one applies the points raised up to here, one quickly discovers that Tesla had found a number of phenomenon that seemed to dispute the classic theories as regards the nature of electrical energy.

Besides the transmission of electrical energy, Tesla while in the midst of the Colorado Springs experiments found a series of other applications for the device. Among these were the use of the "magnifying transmitter" to increase by initially 3 1/3 to 1 and later by as much as 250 to 1 the amount of power (watts) that it used to pump the earth. Tesla's observations on this effect lead him to later write his close friend Robert Underwood Johnson that he had discovered the principles underlying the so-called "fuelless motor". In addition to these claims, Tesla also found that he had developed a unique type of radar and a new method for the broadcasting of "wireless" telegraphy. In short, he had discovered at the dawn of the age of radio broadcasting an entirely different method to transmit radio waves - not through the air, but instead through this very planet itself!

To Nikola Tesla, the "wireless power" system was the core around which he drew the multi-faceted approach to solving the energy crisis. This conclusion was reached by the electrical wizard for three essential reasons. First, the wireless transmission of electrical energy permitted the system to have a transmission capacity unheard of by the more conventional means of the use of electrical transmission lines. If one wished to deliver electrical energy to a remote mountain region or even the furthest portion of the Arctic it would be relatively impossible due to the environmental factors involved. The construction of a world wide network of "magnifying transmitters" solved this problem as quickly as the network was put into operation. All that was required was for the remote region to have a properly constructed receiver that was tuned to the "energy signals" sent by the transmitter. Second, the magnifying capability of the transmitter had to be taken into account. When the earth after a sufficient length of time was properly resonated, the transmitter began to pick up the energy that the earth began to release by nature of its own vibrations. It is the belief of the author that the energy that produced the magnifying effect was due to the collection of the earth-transmitter circuit collecting the ether or "zero-point" energy referred to in the above paragraphs. Hence, the magnifying system was a means to transmit electricity and to generate electricity. Finally as noted above, the system also had other abilities that even today sound fantastic. The system according to Tesla had the ability to act as a type of "over the horizon" radar, a through the earth communications system, a "death ray" and the famed "anti-war machine". Tesla also claimed that the "magnifying transmitter" could control weather and even create man made earthquakes. However after the inability of Tesla to complete the Wardenclyffe prototype commercial site, due in large part to the insufficient monies provided by J.P. Morgan at the cost of 51% control of the patents, the whole concept fell in repute. Between the years 1905 and 1943, no one experimented with the concept other than its discoverer, Nikola Tesla.

Following the completion of the experiments at Colorado Springs that proved the feasibility of the "wireless transmission of electrical energy" to Tesla's satisfaction, the electrical wizard returned to New York City in the early winter of 1899-1900 and was immediately commissioned by his friend and co-editor of Century Magazine, Robert Underwood Johnson, to write an article for the century ending commemorative issue of June, 1900. This article was to become the magnum opus for Tesla's approach to the whole field of energy utilization. This article entitled "The Increasing of Human Energy" was meant by Tesla to explain in philosophical form the knowledge gleaned by the great discoverer over more than a decade of intensive research in the modern miracle called electricity. To start off, Tesla attempted to explain in the language of contemporary physics, the nature of the human condition. To many, the attempt may appear to be too mechanical and devoid of those characteristics that define the human condition; however, this approach often forgets that Tesla was trying to explain the nature of the human mass in as scientific a way as possible. To do this explanation, the categories were divided up into those items which retarded the human mass and those that accelerated the human mass. The major retardant to the human mass was stated to be war. The danger to modern technology was that vital resources were wasted by the constant need to construct modern devices of war such as the battleship and the advent of the age of electricity would increase the waste

and destruction. Tesla wondered if the fact that the increased destructiveness would eventually put an end to war. Tesla's answer was that unfortunately this was not the case. Tesla then proposed that war be fought by a race of specially designed teleautomatons. This type of remote controlled war machine was humanity's only hope to avoid the wastefulness of modern warfare. On the other hand, Tesla also discussed those elements which accelerate the human mass. Here, Tesla firmly believed that the act of invention was the key to human progress. The discoverer of new principles enabled the creation of a vast and uplifting technologies to be invented. Nikola Tesla always had the highest regard for those persons who by the insight and keen observation of their mind opened the door for the discovery of new scientific and/or artistic principles. Such persons were the epitome of the good qualities that composed the human condition.

With the very lengthy philosophical opening completed, Tesla turned to an examination of the energy useage of the present day (circa 1900) and the future (the last half of this century). Tesla's first point was to show the dangers of continuing to utilize fossil fuels. Fossil fuels warned Tesla would pollute the planet and especially its atmosphere. The pollution of the planet's atmosphere would ultimately alter the climate of the globe and especially the more industrialized northern hemisphere. Such a threat would also endanger what is today called by biologist the life sphere or ecosphere. This warning in the last quarter of the twentieth century is becoming quite prophetic. The new science of ecology has in the past thirty years proved with the collection of a rather substantial data base the truth of Tesla's proclinations. The use of fossil fuels has become in many ways an even greater abomination then the electrical prophet forecast at the turn of the century. The key to Tesla's system was the immediate conversion of our technology to energies that were non-polluting. Tesla defined these clean energies as hydroelectric power and solar energy. However, Tesla's categories were much broader than the names for the categories seem to imply. Hydroelectric power was any means used to provide electric energy by the use of water power. These uses included the employing of tides (e.g., the Bay of Fundy), waterfalls, rapids and even exotic technology such as the use of a special fuel cell invented by Tesla at the beginning of this century. Solar energy was divided up into the more traditional solar collectors, solar cells, and the "radiant energy devices that Tesla patented in 1901. To tie this whole energy system together, Tesla intended to make full use of the "wireless" whose testing at Colorado Springs had recently proven to be so successful.

To the concept of a clean energy foundation to our civilization, Tesla added the belief in an energy mix. This concept was the need for each region of the planet to develop to the best of its ability those clean energy resources that were available. To augment the difference between the energy available and the potential energy needs, Tesla proposed the use of the wireless system to broadcast the excess energy of one region to another region that was energy deficient. In this way, Tesla expressed his deep felt belief that the only way to solve the energy problems of the planet was to put an end to the international rivalries of the past few centuries of human existence. In Tesla's energy efficient world, the nation-state had ceased to be the roadblock to human progress that it now unfortunately still appears to be.

TABLE 1.4 THE TESLA "MULTI-FACETED SOLUTION" TO THE "ENERGY CRISIS"

Known Energy Sources	Replacement	Energy Policy Employed
Fossil Fuels	'Magnifying Transmitter'	The solution is based on the employing of the best possible "mix" of clean and renewable energy sources that are most efficient in any given region of the planet. In addition, the wide use of the "wireless power" system to broadcast needed energy to a specific region.
Petroleum		
Coal		
Natural Gas		
Solar Energy	Radiant Energy Machines	
Solar Collector		
Steam Generators		
Nuclear Fuel	Hydroelectric Power	
Traditional		
"Breeder"		

(Source: N. Tesla, "The Increasing of Human Energy" Century Magazine, June, 1900)

The final component in Tesla's "multi-faceted solution" to the energy crisis involved the use of new and traditional clean energy resources. The major part of this segment was the "radiant energy" machines patented by Nikola Tesla in 1901. The energy device was based on the converting of the "magnifying transmitter" to a passive receiver that was tuned to obtain the "zero point energy" that was part of the radiant or heat energy that was sent out in gigantic streams by the Sun. To accomplish this conversion to a passive system, Tesla replaced the coils of the receiver with a special unit whose major part was two large plates that would be able to receive a different potential or "charge". This change in potential would be sent to a specially constructed condenser unit that was capable of holding a charge upon it for a long time without dissipating said charge. In this way, Tesla believed that it would be possible to obtain the energy from the radiant energy of the Sun that the "magnifying transmitter" captured from the earth. Tesla believed that this use of the potential in the energy from the Sun was more efficient than the indirect use of solar energy that is seen in various devices such as solar collectors and solar steam generation plants.

Again, Tesla felt that the maximum efficiency for any device that used solar energy would come from those regions there most abundant in the receiving of this vital energy source. Moreover, the "radiant energy" machines had the capability to be used over a twenty four hour period. This feat was possible due to the fact that the energy collected by the "radiant energy" machine was still present in the earth's atmosphere in large quantities even when the Sun had set or the day was extremely cloudy. The policy of the dispersal of the energy provided by this device would as stated previously be based on the energy mix that would prove best for a particular region.

As we have seen, the planet and especially its largest consumer, the United States, is caught in the midst of an energy crisis. In the beginning of this chapter, we examined the nature of this crisis and learned that the major element in this crisis was the dependence of our advanced technology on natural gas and crude oil. Of all the fossil fuels available to us, these two resources represented the lowest quantities of reserves available to the planet for a continued long term basis. In addition to the need for the consumption of fossil fuels, recent meteorological evidence seemed to indicate that the use of fossil fuels over the next two decades would lead to a massive alteration in the world's weather patterns by the beginning of the next century. This crisis was further complicated by the fact that many experts believed that nuclear power plants were the answer to the energy crisis. Nuclear power plants were extremely dangerous for two important reasons. First, nuclear plants have not yet solved how to successfully dispose of their highly radioactive waste cores. These highly radioactive waste cores serve as an environmental danger to the continuation of the ecosphere. Second, there is the problem of thermal pollution from the water that is employed to cool those parts that are in direct contact with the reactor. The "waste water" is extremely hot when returned to its source and this factor, in turn, contributes to the destruction of various types of fish and plants that are essential to a number of important food chains. These two problems seem to limit the future expansion of nuclear plants that was forecast by many believers in the nuclear age at the beginning of the second half of this century. Thus, the technology as envisioned by the present state of the art

leaned heavily on the development of a sensible solar technology. Such a technology does not appear to be found in the fantastic schemes for solar collector "farms" that have been proposed by many followers of the solar energy. In view of these difficulties, one may ask what technology can be appropriate to successfully solving our present "energy crisis"?

To those who have read the results of the second part of this chapter, the answer is quite obvious - the Tesla "multi-faceted solution". This solution permits the world to develop a technology that not only solves the present energy crisis, but also permits the spreading of the benefits of modern technology to the underdeveloped world without the danger that this spread would add to the climatological threat previously mentioned. Basically, the plan provides for a number of clean energy resources that can be tied together by the use of the Tesla "wireless transmission of electrical energy". In adopting this synergistic plan of action, the planet would not only free itself of a growing dependence on fossil fuels, but also provide a means for international cooperation on a global scale. Such an operation can not hurt the cause of international peace and justice that Nikola Tesla so proudly proclaimed during his lifetime.